Large Language Models in Document Intelligence: Survey

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1. Introduction

Document intelligence focuses on the automatic extraction, understanding, and analysis of business documents. Recent advances in Large Language Models (LLMs) and Vision-Language Models (VLMs) have reshaped traditional document analysis workflows into unified reasoning pipelines. In this survey, we examine over 300 research publications between 2021 and 2025.

2. Architectural Paradigms

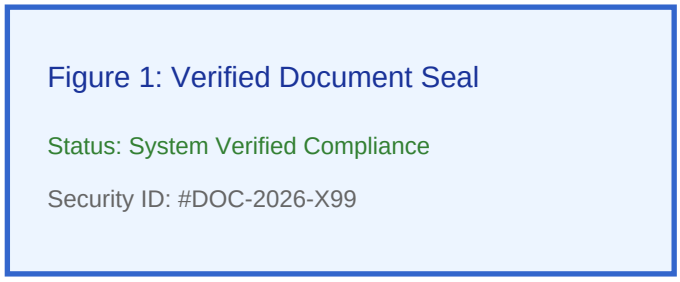
Modern systems are categorized into pipeline-based approaches and end-to-end multimodal architectures. Pipeline systems decompose tasks into Document Layout Analysis (DLA), Table Structure Recognition (TSR), and Optical Character Recognition (OCR), followed by Retrieval-Augmented Generation (RAG). Meanwhile, multimodal LLMs process document page images directly, preserving non-linear reading order and spatial visual structures.

3. Empirical Performance and Evaluation

Table 1: Quarterly Financial and Metric Comparison (Fiscal Year 2025)

Quarter	Revenue ($M)	Expenses ($M)	Net Profit ($M)	Accuracy (%)
Q1-2025	120.5	85.2	35.3	92.4
Q2-2025	145.0	92.0	53.0	94.8
Q3-2025	180.2	105.1	75.1	96.2
Q4-2025	210.0	118.5	91.5	97.5

Observations from Table 1 demonstrate consistent quarter-over-quarter revenue expansion, rising from $120.5M in Q1 to $210.0M in Q4. Net profit grew by $56.2M across the full year, with document extraction accuracy reaching an optimal 97.5% in Q4.

4. Future Challenges and Conclusions

Key remaining challenges include handling ultra-long context documents, maintaining high-fidelity spatial alignment for complex non-standard templates, and preventing hallucination in high-stakes financial and regulatory domains. The integration of deterministic tool verification with multimodal LLMs marks a decisive step toward trustworthy enterprise document intelligence.